MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE, NEGATIVE GRID
TYPE 8403

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic-metal
See figure 1
Mounting position: Any
Weight: 2.5 ounces (71 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Fl	Ef	Eb	epy	Ec	ib	Ib	Ic	tp
Unit:	GHz	V	kVdc	kv	Vdc	a	mAdc	mAdc	µs
Anode pulsed osc or amp:									
Maximum:	3.0	6.6	---	3.5	-150	3.0 (see note 3)	30	10	6
Minimum:	---	6.0 (see notes 1 and 2)	---	---	---	---	---	---	---
Grid pulsed osc or amp:									
Maximum:	3.0	6.6	2.5	---	-150	3.0 (see note 3)	30	10	6
Minimum:	---	6.0	---	---	---	---	---	---	---
CW:									
Maximum:	2.5	6.6	---	---	-150	---	150	---	---
Minimum:	---	6.0	---	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3	1,000	---	Adj	---	100	---	---

ABSOLUTE RATINGS:

Parameter:	Pg	tk	TE	T (anode shank)	Du	Pp	Barometric pressure, reduced
Unit:	W	sec	°C	°C (see note 5)	---	W	mmHg
Anode pulsed osc or amp:							
Maximum:	2 (see note 4)	---	250	250	0.01	58	54
Minimum:	---	60	---	---	---	---	---
Grid pulsed osc or amp:							
Maximum:	2 (see note 4)	---	250	250	0.01	33	54
Minimum:	---	60	---	---	---	---	---
CW:							
Maximum:	2	---	250	250	---	---	---
Minimum:	---	60	---	---	---	---	---
TEST CONDITIONS:	---	300	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	6							
1231	Pulsing emission	-	(eb - ec)/is = 10 a; tp = 3 µs (max); prr - 600 (max)	0.65	II	etd	---	200	v
1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	R	50	---	Meg
1301	Heater current	-		0.65	II	If	1.20	1.40	A
1261	Electrode voltage (1) (grid)	7		0.65	II	Ec	-4.0	-9.0	Vdc
1266	Total grid current	7		0.65	II	Ic	---	-8.0	µAdc
	Quality conformance inspection, part 2								
1261	Electrode voltage (2) (grid cutoff)	-	Ec/Ib = 1.0 mAdc; Eb = 1,000 Vdc	---	---	Eco	---	-30	Vdc
---	High voltage hold off	-	Eb = 2,500 Vdc; Ec = -130 Vdc	---	---	Ib	---	1.0	mAdc
1331	Direct-interelectrode capacitance	8,9	No voltages applied	---	---	Cin Cgp Cout	7.5 2.95 ---	9.0 3.25 0.06	pF pF pF
---	Resonance	10	No voltages applied; TA = 25° ±5°C	---	---	---	---	---	---
1236	Power oscillation (1) (pulse)	11	F = 1.1 GHz +50 MHz; Ec = -80 Vdc; Ebb = 3.0 kVdc; Ef = 6.3 vac; ec = 105 v (grid pulse)	---	---	Po	2.0	---	W
	Quality conformance inspection, part 3								
(A) ---	Life test	16	Group C; heater standby; t = 500 hours; Ef = 6.3 V	---	---	---	---	---	---
---	Life-test (1) end point	16		---	---	Δib	---	25	%
1002	Barometric pressure, reduced	13	Pressure - 54 mmHg (max); voltage = 1,000 Vac; TA = 30° ±10°C	---	---	---	---	---	---
1031	Variable frequency vibration	8,12, 13	F = 55 to 500 Hz; 15 G peak (min); Ec/Ib = 10 mAdc; Rp = 10,000 ohms; Ebb = 300 Vdc	---	---	Ep	---	250	mVac
1041	Shock	8,13, 15	60 G peak (min); duration = 11 ±2 ms; no voltages applied	---	---	---	---	---	---
---	Torque	8,13, 14	No voltages applied	---	---	---	---	---	---
---	Shock and torque-test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	µAdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
1236	Power oscillation (2) (pulse)	8,13, 17	F = 3.0 GHz (min); epy = 3.5 kv; Ec = -1.5 Vdc (min); Rg/Ib = 15 mA (max); Ef = 5.8 Vac	---	---	Po	5.0	---	W
---	Power gain	8,13, 18	F = 1.1 GHz ±50 MHz; Ebb = 2.2 kVdc (min); Ecc = -50 Vdc; tp = 3 µs (min); Du = 0.002 (min); pd = 400 w	---	---	po	1.8	---	kw
1236	Power oscillation	8,13, 19	F = 2.5 GHz ±100 MHz; Ef = 5.3 Vac; Eb = 1.0 kVdc; Ib = 150 mA	---	---	Po	22	---	W
---	Life test (2)	13, 20	Ef = 6.3 Vac; F = 5 kHz ±10%; Ebb = 1.0 kVdc; Eg and Rk/Ik = 125 mAdc (min); Ic = 45 mAdc (min); TA = 250°C (min); t = 100 hours	---	---	---	---	---	---
---	Life-test (2) end point:								
Ⓐ 1231	Pulsing emission	-	(eb = ec)/is = 8 a	---	---	eid	---	200	v

NOTES:

1. The transit time heating effect of the cathode shall be compensated by a reduction in heater voltage after dynamic operation of the tubes has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.
2. Where emphasis is placed on long and reliable life, the filament voltage can safely be lowered to 6.0 volts, provided the line voltage is regulated better than ±2 percent.
3. Regulation, or series-anode-supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.
4. The maximum instantaneous peak grid-cathode voltage shall be within the range of +250 to -750 volts.
5. Sufficient conduction, convection, and forced-air cooling shall be provided to limit the envelope and anode shank temperatures to the specified maximum value under all operating conditions. Reliability will be seriously impaired if this maximum is exceeded. Where emphasis is placed on long and reliable life, lower temperatures shall be used.
6. All test listed under quality conformance inspection, part 1, shall be performed at the conclusion of the holding period.
7. Sufficient conduction, convection, and forced-air cooling shall be used in all electrical tests involving application of heater voltage to maintain the anode shank and seal temperatures within the specified maximum values.
8. Other tube contact configuration may be used provided the tube contact area remains unchanged, and the socket, jig, or cavity gives equal performance. The socket, holding fixture, or cavity shall be rigidly secured to the test platform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

9. Measure in socket in accordance with Defense Electronics Supply Center Drawing No. 68017.

10. Grid-anode resonance. Cavity shall resonate at 4,980 MHz ±10 MHz in accordance with Defense Electronics Supply Center Drawing No. 67032.

    Grid-cathode resonance. Cavity shall resonate at 1,377 MHz ±2 MHz in accordance with Defense Electronics Supply Center Drawing No. 67033.

    When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	Cgp	Cgk	Fgp	Fgk
1	2.95	7.0	1,465	1,461
2	2.95	7.0	1,425	1.421
3	3.25	9.0	1,380	1,440
4	3.25	9.0	1,340	1,400

11. The applied voltage pulse shape shall be tp = 0.5 µs ±10 percent, tr = 0.05, and tf = 0.1 µs maximum. The pulse repetition rate shall be adjusted so that Du = 0.001 ±5 percent with the above specified pulse length. Test in cavity MCC Model T-104, or equivalent. The cavity and circuit connections are shown in figures 2 and 3. The grid drive is measured with the heater and bias voltage "on" and the anode voltage "off". Under these conditions the peak positive voltage with respect to the cathode should be egk = 25 volts maximum.

12. The tube shall be mounted in socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximately logarithmic progression and shall require 4 minutes minimum, 6 minutes maximum to traverse the range. Each tube shall be vibrated for 30 minutes in each position -X and +Z as shown on figure 1, except that if the cumulative result of test on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltage in one position, subsequent measurements need to be taken only in the position giving the higher reading. The voltages specified herein shall be applied to the tube during vibration. The value of the alternative voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range, shall have the ability to measure, with an error of less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output Ep, shall not exceed the limit specified herein at any point in the sweep frequency range during the last complete cycle of cycling vibration.

(A) 13. This test shall be performed during initial production and then annually thereafter. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The annual double sampling plan shall be reinstated after three consecutive samples have been accepted.

14. Torque test shall be performed as follows:

    (a) The tube shall be held securely at the cathode connection. A force of 5 inch-pounds shall be applied to the heater cup without perceptible shock. This test may be made by applying the force at right angles to the inside of the cup at a point 0.109 ±0.016 inch from the cathode end of the tube. An approved equivalent method may be used. The heater cup shall not loosen or short circuit on the cathode connection. This part of the test shall not be required if the space between the heater cup and the cathode sleeve is completely filled with insulating material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

(b) A torque of 15 inch-pounds shall be applied between anode and cathode without shock.

(c) A torque of 40 inch-pounds shall be applied between anode and grid without shock.

15. Test in jig in accordance with Drawing 280-JAN. Each tube shall be subjected to a total of 15 shocks; that is, 5 shocks in each position, +Y, +Z, and -Z, in any sequence, as shown in figure 1.

16. At zero hours of life test, establish the drive conditions necessary to obtain a peak anode current of 3a (minimum) with an anode voltage of 1,000 Vdc and a bias of -40 Vdc. The pulse width of the modulator shall be 2 µs (minimum) and the duty shall be 0.0025 (maximum). With the drive level determined at zero hours, check the anode current at the end of life. Maximum allowable drop in anode current shall be 25 percent.

17. The applied voltage pulse shape shall be measured with a noninductive resistor of 1,150 ohms ±2 percent inserted in place of the tube. The pulse shape shall be tp = 3.0 µs ±10 percent, tr = 0.4 µs maximum, tf = 0.7 µs maximum. The pulse repetition rate shall be adjusted so that Du = 0.0025 ±5 percent with the above measured pulse length. Test in cavity in accordance with Drawing 279-JAN. The heater contact of the cavity may be lengthened as required and the anode flange "M" removed to attain proper oscillation. The cavity shall be connected to a load and a VSWR of less than 1.5. The oscillator output coupling and the grid or cathode resistor may be adjusted for maximum power output.

18. Test shall be conducted in power amplifier cavity as shown in figure 4. Driving power is defined as the net power delivered to the amplifier cavity input terminals and the reflected power shall be subtracted from the incident power to obtain the net driving power. The output tuning shall be adjusted for maximum power output.

19. Test is to be made in cavity in accordance with Drawing 160-JAN. The position of the cathode line and the heater contact are changed to permit proper contact as required by the tube dimensions. The anode flange "M" may be removed to obtain proper oscillation. The cavity shall be connected to a standard load with a VSWR of less than 1.5. The output coupling from the oscillator and grid or cathode resistor or both may be adjusted for maximum power output.

20. Test tubes in circuit in accordance with figure 5, or equivalent.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 17, 80
DSA - ES

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2988)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

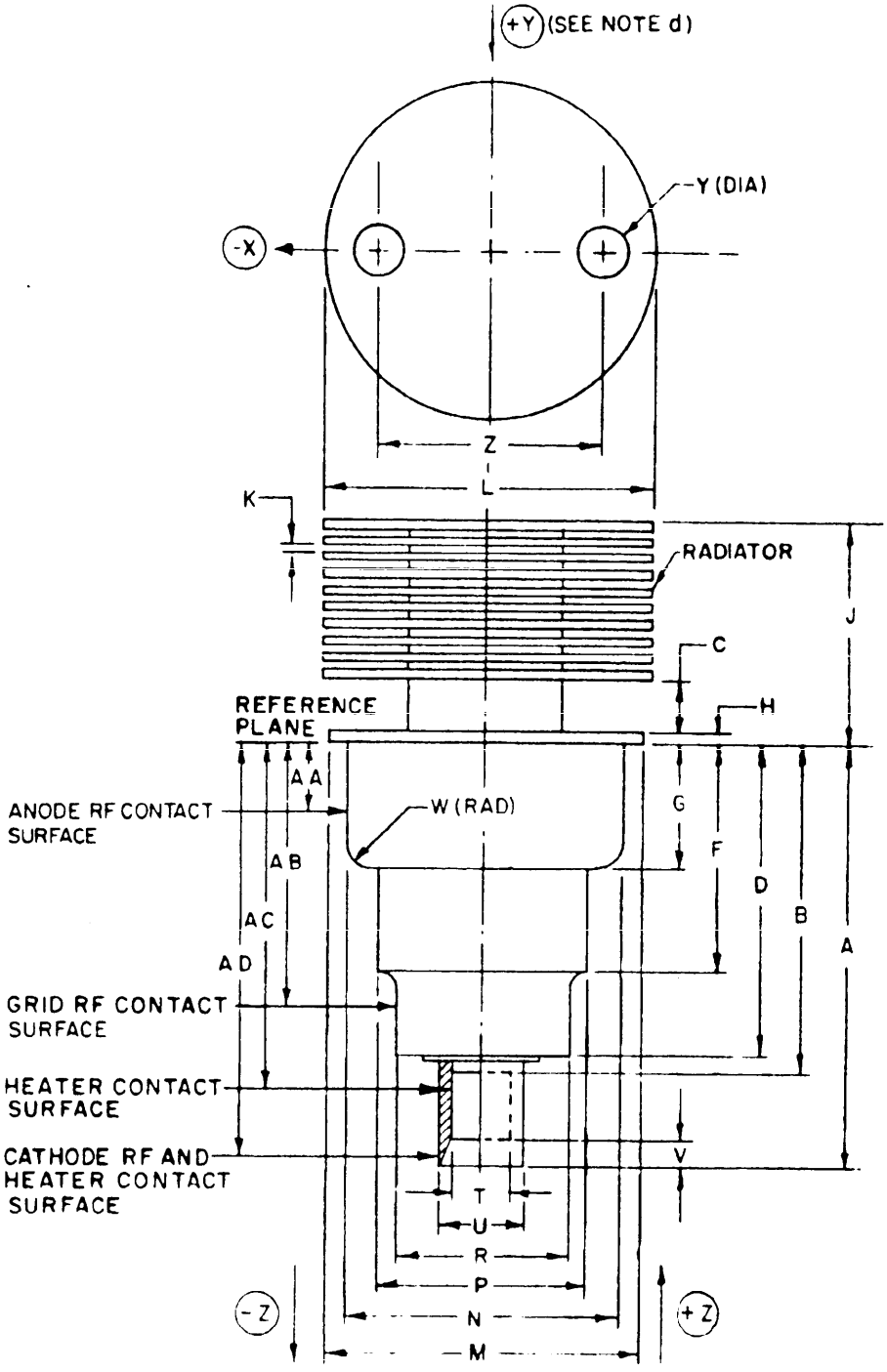


FIGURE 1. Outline drawing of electron tube type 8403.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	1.500 (38.10)	1.560 (39.62)	
B	---	1.214 (30.84)	
D	1.125 (28.58)	1.165 (29.59)	
F	.800 (20.32)	.840 (21.34)	
G	.462 (11.73)	.477 (12.12)	
N	1.025 (26.04)	1.035 (26.29)	b
R	.655 (16.64)	.665 (16.89)	b
T	.213 (5.41)	.223 (5.66)	
U	.315 (8.00)	.325 (8.26)	c
(A) Quality conformance inspection, part 3 (SEE NOTE a)			
H	---	.040 (1.02)	
C	.125 (3.18)	.185 (4.70)	
J	.766 (19.46)	.826 (20.98)	
K	.025 (.64)	.046 (1.17)	
L	1.234 (31.34)	1.264 (32.11)	
M	1.180 (29.97)	1.195 (30.35)	
P	.752 (19.10)	.792 (20.12)	
V	---	.086 (2.18)	
W	---	.100 (2.54)	
Y	.105 (2.67)	.145 (3.68)	
Z	.650 (16.51)	.850 (21.59)	

Dimensions for electrode contact surfaces		
Ltr	Minimum	Maximum
AA	.035 (.89)	.361 (9.17)
AB	1.021 (25.93)	1.101 (27.97)
AC	1.219 (30.96)	1.413 (35.89)
AD	1.160 (29.46)	1.500 (38.10)

NOTES:

a. Dimensions shall be checked during the initial production and annually thereafter. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. Annual double sampling plan shall be reinstated after three consecutive samples have been accepted.

b. The total indicated runout of the anode and grid contact surfaces with respect to the cathode contact surface will not exceed .020 (.51 mm).

c. The total indicated runout of the cathode contact surface with respect to the heater contact surface will not exceed .012 (.30 mm).

d. -X, +Y, +Z and -Z indicate direction of the applied shock and vibration in relation to the tube configuration.

FIGURE 1. Outline drawing of electron tube type 8403 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

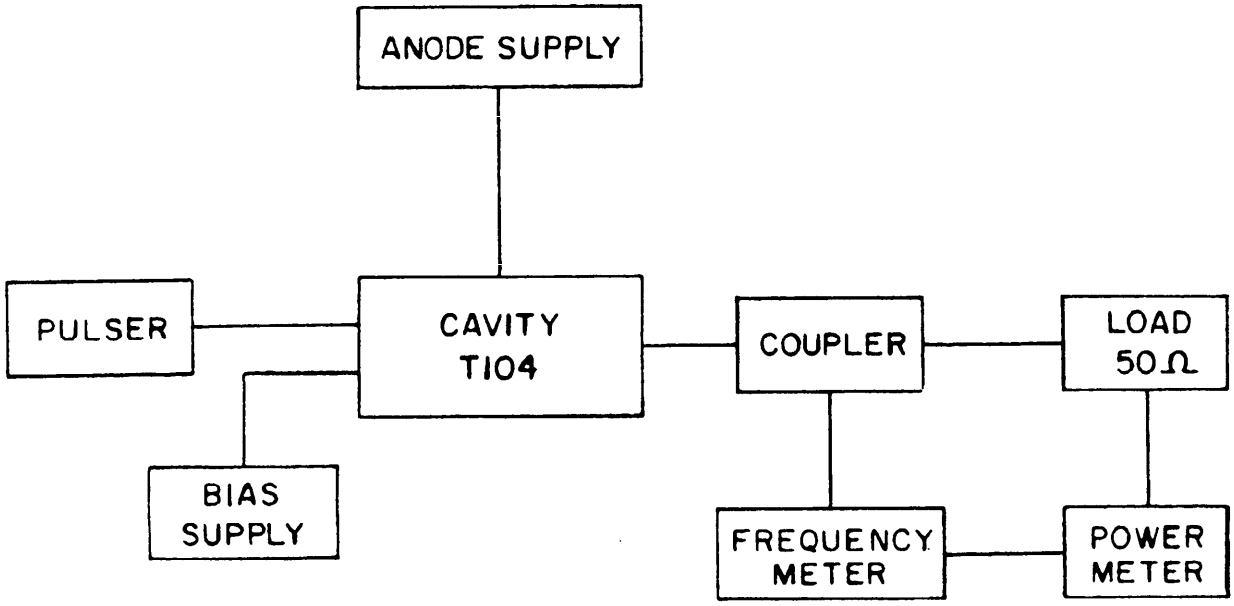


FIGURE 2. Test cavity hook-up.

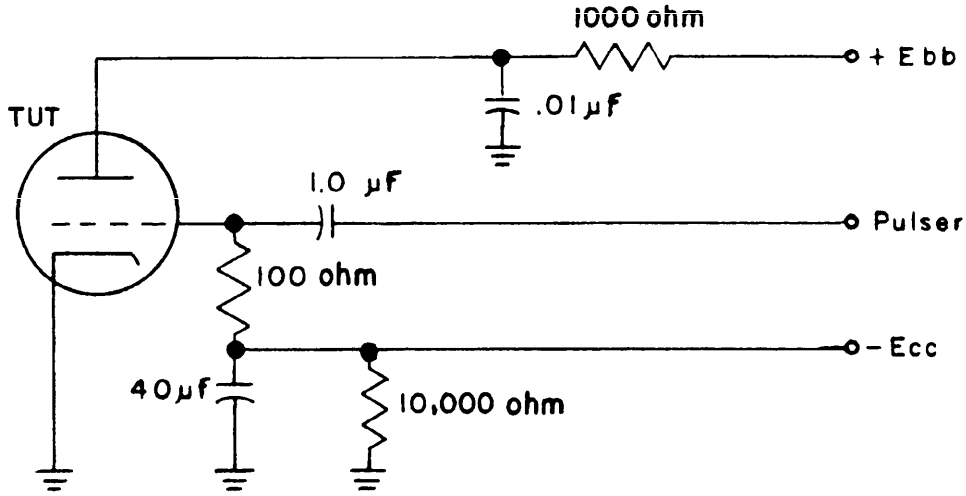
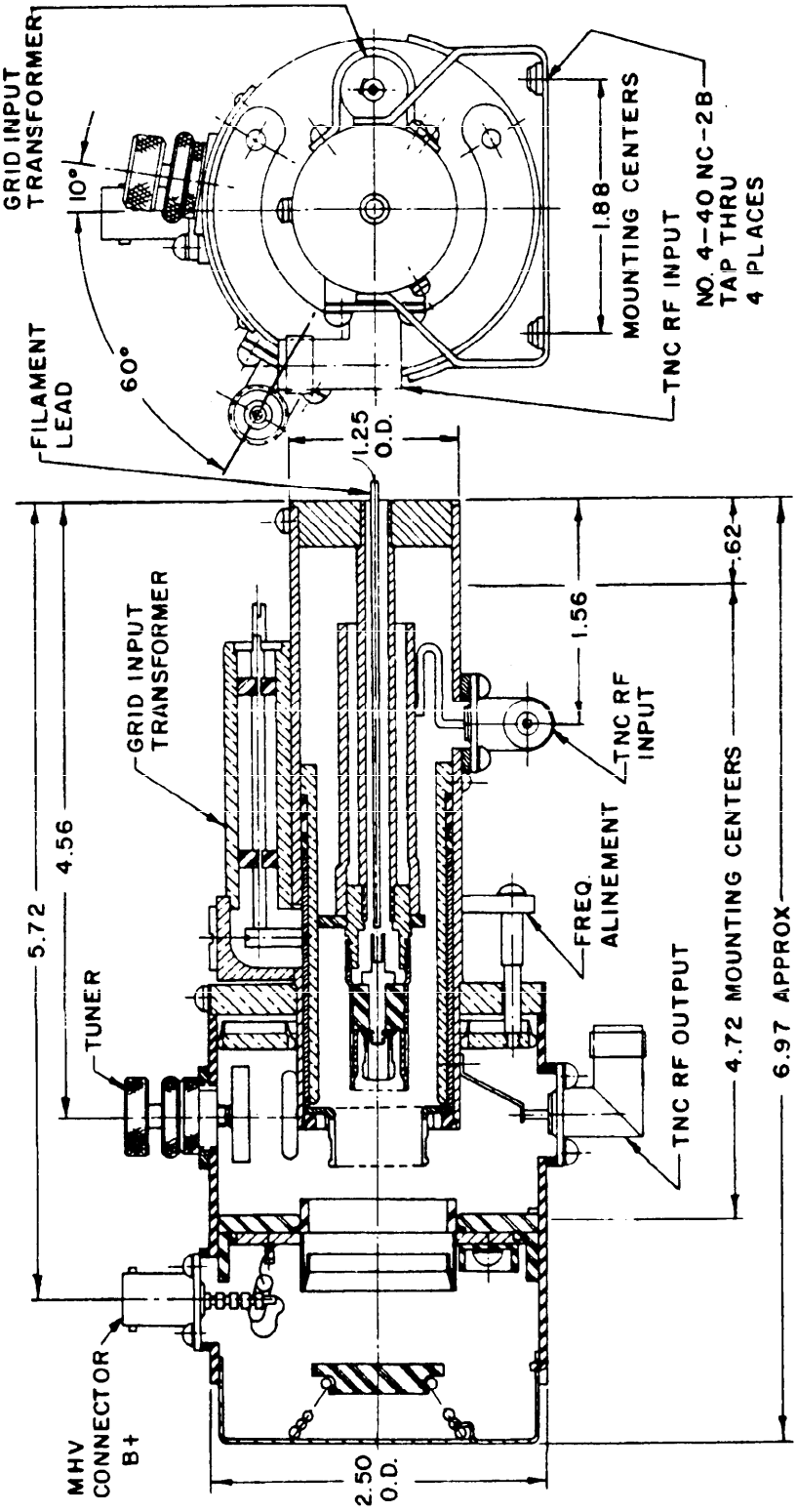


FIGURE 3. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.



INCHES	MM
.62	15.75
1.25	31.75
1.56	39.62
1.88	47.75
2.50	63.50
4.72	119.89
4.56	115.82
5.72	145.29
6.97	177.04

NOTES:
a. Dimensions are in inches.
b. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
c. Unless otherwise specified, tolerance is ±.02 (.51 mm).
d. All dimensions are for reference only.

FIGURE 4. Pulse amplifier cavity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

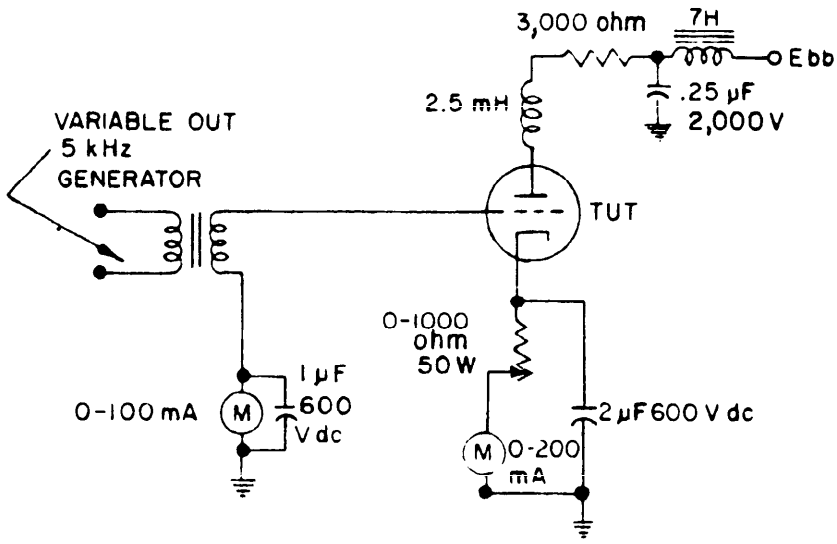


FIGURE 5. 5 kHz life-test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:14Z
Check the source to verify that this is the currentversion before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAI
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:14Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202640-15T15:14Z
Check the source to verify that this is the current version before use.